

Chapter 12: Ectopic Pregnancy — Revision Notes

Williams Obstetrics 26e, pp. 574–613 of the PDF. High-yield numbers are in **bold**. The one table is transcribed from the PDF image (it is missing from the extracted chapter text). All 9 figures are included from the PDF (Figure 12-4 spans two pages and is shown once).

Big picture: Any implantation outside the endometrial cavity is ectopic. Rates are **~1.4–1.5%** of pregnancies, and ectopic pregnancy causes **3% of pregnancy-related deaths**. Serum **β -hCG + transvaginal sonography** allow early diagnosis, so most women can be treated with **methotrexate** or **laparoscopy** and keep their fertility.

1. Tubal Pregnancy

Classification

- **~95%** of ectopics are **tubal**.

Site	Share of tubal ectopics
Ampulla	70% (most common)
Isthmus	12%
Fimbria	11%
Interstitial	2%

- **Nontubal (~5%)**: ovary, peritoneal cavity (abdominal), cervix, cesarean scar.
- **Heterotopic** = an intrauterine and an ectopic conceptus together.
- Management depends on viability, gestational age, maternal health, fertility wishes, surgeon skill and resources.
- **D-negative women with any ectopic get anti-D immunoglobulin: 50 or 120 μ g IM in the 1st trimester; 300 μ g later.**

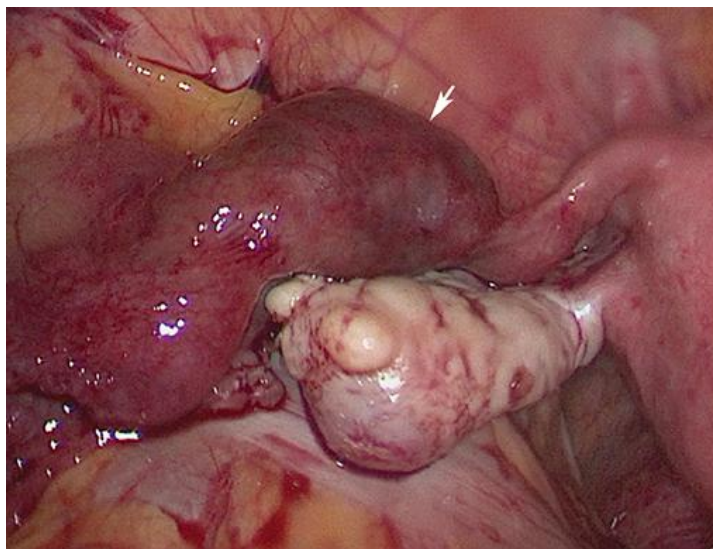


Figure 12-1. Laparoscopic view of an unruptured ampullary tubal pregnancy (arrow), the most common site (70%).

Risk factors

- **Abnormal tubal anatomy** underlies most cases.
- **Highest risk: prior tubal surgery** (for a previous ectopic, fertility restoration or **sterilization**).
- **Prior ectopic** → **~10% recurrence**.

- **Prior salpingitis:** one episode → up to **9%** later ectopic. Peritubal adhesions (salpingitis, appendicitis, endometriosis).
- **Infertility and ART.**
- **Smoking.**
- **Contraceptive failure** with methods that block intrauterine implantation better than tubal: **tubal sterilization, IUDs, progestin-only contraceptives.** (The absolute number of ectopics still falls with any contraception.)

Pathogenesis and potential outcomes

- The tube **lacks a submucosa** → the ovum burrows through the epithelium into the muscularis, which trophoblast invades.

Outcome	Features
Tubal rupture	Usually spontaneous (occasionally after coitus or bimanual exam); hemorrhage often persistent and life-threatening
Tubal abortion	Passes out the fimbriated end; bleeding may stop, or blood pools in the cul-de-sac ; occluded fimbria → hematosalpinx ; rarely reimplants → abdominal pregnancy
Spontaneous failure	Death and reabsorption; now detected more often with sensitive hCG assays

	Acute ectopic	Chronic ectopic
Frequency	More common	Less common
β-hCG	High , grows fast	Negative or low and static (trophoblast dies early)
Rupture	Greater risk , early	Late, if at all
Presentation	Timely diagnosis	Persistent complex pelvic mass (often found on imaging)

Clinical manifestations

- **Differential of early-pregnancy pain:** miscarriage, infection, leiomyoma degeneration, round-ligament pain; ovarian mass (hemorrhagic, ruptured, **torsed**); appendicitis, renal stone, cystitis, gastroenteritis.
- **Initial work-up:** urine β-hCG, urinalysis, Hb/Hct (**CBC** if infection possible). Positive urine test + pain or bleeding → **serum β-hCG**.
- **Classic triad:** amenorrhea → **pain + vaginal bleeding**. Before rupture, signs are often subtle or absent.
- **Rupture:** severe, **sharp, stabbing or tearing** lower abdominal/pelvic pain.
 - **Neck or shoulder pain on inspiration** = diaphragmatic irritation from hemoperitoneum.
 - **Vertigo, syncope** = hypovolemia.
- Most women have some spotting or bleeding; profuse bleeding suggests incomplete abortion but can occur with tubal pregnancy.
- **Exam:** abdominal tenderness; adnexal mass/tenderness on a **limited, gentle** bimanual exam (avoid iatrogenic rupture); uterus may be slightly enlarged.
- **Vital signs:** moderate bleeding may cause no change, a slight BP rise, or a **vasovagal response (bradycardia + hypotension)**. BP falls and pulse rises only with significant hypovolemia.
- **Labs:** initial Hb/Hct may be only slightly low → a **falling trend over several hours** is more useful. **Leukocytosis up to 30,000/μL** in ~half of ruptured ectopics.
- **Decidual cast:** sloughed endometrium shaped like the cavity; also occurs with miscarriage → send tissue for histology. **No sac or no villi** → **ectopic is still possible**.

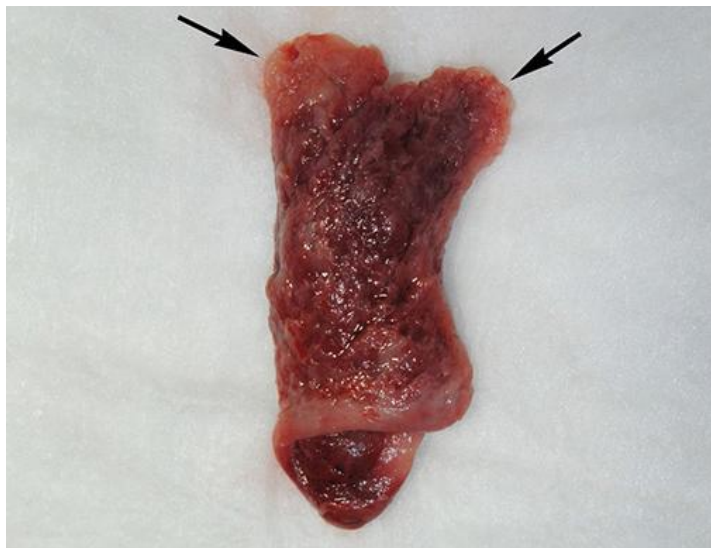


Figure 12-2. A 7-cm decidual cast passed with a tubal ectopic; arrows mark the decidua that lined the cornua. A cast alone does not prove an IUP.

2. Tubal Pregnancy Diagnosis

- **Evidence of rupture** → **prompt surgery**. Hemodynamically stable women without a clearly located pregnancy → β -hCG, TVS $\pm$ surgery.
- **Trade-off**: strategies that maximize ectopic detection may **interrupt a normal IUP**; those that protect an IUP may **delay** ectopic diagnosis. The patient's wishes for the pregnancy guide the balance.

β -hCG

- Immunoassays detect the β -subunit. Detection limits: **urine 20–25 mIU/mL**; **serum ≤ 5 mIU/mL**.
- Assays vary by **5–10%** → serial values from the **same laboratory**.
- **Discriminatory threshold**: the level above which a normal IUP should be seen on TVS: **≥ 1500 or ≥ 2000 mIU/mL** (institution-specific). With live IUPs, a sac was seen **99%** of the time above **3510 mIU/mL**.

Transvaginal sonography

Pregnancy of unknown location (PUL)

- No IUP or ectopic seen on TVS. Usually one of:
 1. **Failing IUP**
 2. **Recent completed abortion**
 3. **Early IUP**
 4. **Ectopic pregnancy**
- Without clear ectopic evidence → **repeat β -hCG at 48 h** (avoids unnecessary MTX and harm to an early IUP).

β -hCG pattern over 48 h	Interpretation
Rise $\geq 53\%$ (conservative minimum 35%)	Consistent with a normal IUP (same expected rise with multiples)
Rapid fall (Table 11-1)	Failed IUP / failing PUL (fall 35–50% at 48 h , 66–87% at 7 days for starting values 250–5000)
Anything else	Suspicious for ectopic

- **⚠ One third of ectopics rise $\geq 53\%$ in 48 h**. About **half** of ectopics have falling and half rising levels.
- **⚠ A resolving ectopic can rupture even with falling or low β -hCG** (hCG may not reach the circulation when the vascular connection is partly disrupted) → follow β -hCG **until negative**.

- After the first two values, repeat every **2–7 days** (more often if higher risk), ± repeat TVS.
- **D & C** gives a faster diagnosis but may interrupt a normal IUP; repeat TVS first.

Endometrial findings

Finding	Timing / meaning
IUP gestational sac	4½–5 weeks
Yolk sac	5–6 weeks
Fetal pole with cardiac activity	5½–6 weeks (transabdominal: slightly later)
Trilaminar endometrial pattern	Characteristic of ectopic: specificity 94%, sensitivity 38%
Endometrial stripe <8 mm in a PUL	No normal IUPs had a stripe this thin
Pseudogestational sac	Fluid between the endometrial layers, central , conforms to cavity shape → ↑ ectopic risk
Decidual cyst	Anechoic area within the endometrium, remote from the canal, often at the endometrial–myometrial border (early decidual breakdown)
Intracavitary sign (IUP)	Anechoic sac placed eccentrically within one endometrial layer

- ACOG: **be cautious diagnosing an IUP without a definite yolk sac or embryo.**



Figure 12-3. Pseudogestational sac: a central anechoic collection in the cavity with a trilaminar endometrium, common with ectopic pregnancy.

Adnexal findings

- Diagnosis rests on an **adnexal mass separate from the ovary.**

Appearance	Frequency
Complex (inhomogeneous) adnexal mass	60%
Hyperechoic tubal ring around an empty sac ("bagel")	20%
Obvious extrauterine sac with yolk sac/embryo (diagnostic)	13%

- **"Ring of fire"** on color Doppler = peripheral placental flow. A **corpus luteum cyst** can look similar.
- Not every adnexal mass is an ectopic → integrate with clinical data.

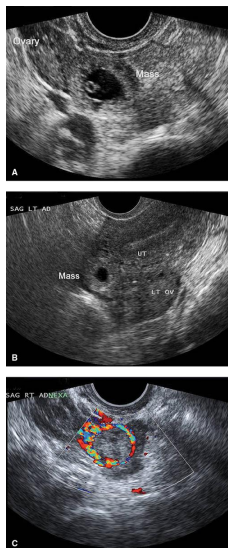


Figure 12-4. Tubal ectopic on TVS, separate from the ovary: (A) extrauterine sac with yolk sac, (B) empty sac with hyperechoic tubal ring, (C) inhomogeneous mass with color-Doppler "ring of fire".

Hemoperitoneum

- Fluid first collects in the **retrouterine cul-de-sac**, then surrounds the uterus, then tracks up the **paracolic gutters to Morison pouch**.
- **Fluid in Morison pouch appears only after ~400–600 mL.**
- **Peritoneal fluid + adnexal mass + positive pregnancy test = highly predictive** of ectopic. Mimic: **malignant ascites**.
- **Culdocentesis** (if no ultrasound): tenaculum pulls the cervix up toward the symphysis; **18-gauge needle** through the **posterior fornix**.
 - No fluid = only an unsatisfactory tap.
 - **Nonclotting** bloody fluid / old clot fragments → **hemoperitoneum**.
 - Blood that **clots** → vessel puncture or brisk bleeding from rupture.

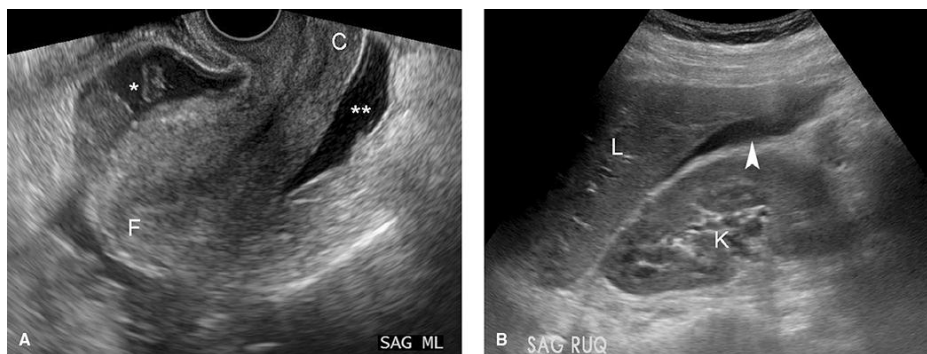


Figure 12-5. Hemoperitoneum: (A) blood in the posterior (**) and anterior (*) cul-de-sac; (B) fluid in Morison pouch (arrowhead), seen only with $\geq 400\text{--}600\text{ mL}$.

Serum progesterone

- Used by some when β -hCG and TVS are inconclusive (not Parkland practice). One value suffices.

Level	Meaning
<6 ng/mL (<20 nmol/L)	Nonviable pregnancy (pooled specificity 98%)
>25 ng/mL	Live IUP , excludes ectopic with 97% sensitivity
10–25 ng/mL	Most ectopics → limited utility ; cannot locate the pregnancy

Endometrial sampling

- Ectopic endometrium lacks villi: **decidual reaction 42%, secretory 22%, proliferative 12%**.

- Without histological exclusion of a miscarriage, a presumed ectopic diagnosis is **wrong in 27%** → some do **D & C before MTX** (weigh against MTX's limited risks).
- **Pipelle biopsy/aspiration is inferior** to curettage. **Frozen section** of curettings identifies products of conception in **95%**.

Laparoscopy

- Reliable diagnosis in most cases and allows immediate **definitive surgery**.

3. Medical Management

Regimen options

- **Medical therapy is preferred when feasible.** Disqualifiers: **ruptured tube** and **drug contraindications**. Also needs access to emergency care and commitment to follow-up labs.
- **Methotrexate (MTX):** folic acid antagonist; binds **dihydrofolate reductase** → blocks dihydrofolate → tetrahydrofolate → stops purine/pyrimidine synthesis → arrests DNA, RNA and protein synthesis. Kills rapidly dividing **trophoblast**; can harm **GI mucosa, bone marrow, respiratory epithelium**.
- **Pretreatment labs:** **serum creatinine** (renally cleared), **CBC and LFTs** (hepato- and myelotoxic), **blood type and Rh**. All but typing are repeated before extra doses.
- **Avoid during treatment:**
 1. **Folic acid** supplements (compete at DHFR)
 2. **NSAIDs** (↓ renal flow, delay excretion)
 3. **Alcohol** (liver enzyme rise)
 4. **Sunlight** (MTX dermatitis)
 5. **Sexual activity** (rupture)
- **Teratogen:** MTX embryopathy = **craniofacial and skeletal abnormalities + FGR**. Excreted in **breast milk**.

Table 12-1. Medical Treatment Protocols for Ectopic Pregnancy

	Single Dose	Multidose
Dosing	One dose; repeat if necessary	Up to four doses of both drugs until serum β -hCG declines by 15%
Medication Dosage: Methotrexate	50 mg/m² BSA (day 1)	1 mg/kg, days 1, 3, 5, and 7
Medication Dosage: Leucovorin	NA	0.1 mg/kg days 2, 4, 6, and 8
Serum β-hCG level	Days 1 (baseline), 4, and 7	Days 1 (baseline), 3, 5, and 7
Indication for additional dose	If serum β -hCG level does not decline by 15% from day 4 to day 7 . Less than 15% decline during weekly surveillance	If serum β -hCG level declines <15%, give additional dose; repeat serum β -hCG in 48 hours and compare with previous value; maximum four doses
Surveillance	Once 15% decline achieved, then weekly serum β -hCG levels until undetectable	(same)

Methotrexate Contraindications

- MTX sensitivity
- Tubal rupture
- Breastfeeding
- Intrauterine pregnancy
- Peptic ulcer disease

- Active pulmonary disease
- Immunodeficiency
- Hepatic, renal, or hematologic dysfunction

BSA = body surface area; β -hCG = β -human chorionic gonadotropin; MTX = methotrexate; NA = not applicable.
From ACOG, 2019c; American Society for Reproductive Medicine, 2013.

	Single-dose	Multidose
Pros	Simpler, cheaper, less monitoring (Parkland's choice)	Some studies show higher success
Leucovorin	No	Yes (folinic acid, buffers toxicity)

- Given **IM**; observe **30 min** after injection. Overall tubal resolution **~90%**.

Patient selection

- **Best candidate: asymptomatic, motivated, compliant.**
- Predictors of success: **low initial β -hCG** (the **best single predictor** with single-dose MTX), **small mass, no cardiac activity**.

Initial β -hCG (mIU/mL)	Single-dose MTX failure
<1000	1.5%
1000–2000	5.6%
2000–5000	3.8%
5000–10,000	14.3%

- Mass **≤ 3.5 cm** → **93% success** vs 87–90% if larger. **≤ 4 cm without cardiac activity** is suitable. **Cardiac activity** → **87% success**.

Side effects

- Usually minimal; resolve **3–4 days** after stopping. **Liver 12%, stomatitis 6%**, gastroenteritis 1%; rare marrow depression.
- **"Separation pain" in 65–75%**, starting several days after MTX; usually mild → analgesics. **20%** need clinic/ED evaluation to exclude rupture.
- **Ovarian reserve is not reduced.**
- **Delay conception ≥ 3 months** after MTX (persists in tissues), although conception within 6 months looked reassuring in one study.

Surveillance

- **Single dose:** β -hCG on **days 4 and 7** (it may rise between days 1 and 4). If the fall from **day 4 to 7 is $<15\%$** → recheck CBC, creatinine, LFTs → **second dose**; that day becomes the new day 1. Needed in **~20%**.
- **Multidose:** MTX 1 mg/kg alternating with leucovorin 0.1 mg/kg; check β -hCG after each pair; $\geq 15\%$ fall expected; **max 4 doses**.
- Once **$\geq 15\%$ fall** → **weekly β -hCG until undetectable**.
- **Mean time to resolution (<15 mIU/mL) 34 days; longest 109 days.**

4. Surgical Management

Options

- Discuss fertility plans; the **other tube can be ligated or removed** if sterilization is desired.
- **Laparoscopy is preferred** unless the woman is **hemodynamically unstable**: same subsequent IUP and patency rates as laparotomy, less infection, adhesion and VTE, faster recovery. Experienced surgeons can

manage hemoperitoneum laparoscopically, but **pneumoperitoneum ↓ venous return and cardiac output** (important with hypovolemia).

	Salpingectomy	Salpingostomy
Use	Ruptured or unruptured	Small unruptured pregnancy
Technique	Bipolar device across the uterotubal junction , then along the mesosalpinx	10–15-mm linear incision on the antimesenteric border ; products extruded or flushed out with high-pressure irrigation; bleeding points coagulated; incision left unsutured (secondary intention)
Persistent trophoblast	Rare	5–15%
Follow-up	No routine β -hCG if asymptomatic	Weekly β-hCG

- **Normal contralateral tube → no fertility difference:**
 - **ESEP:** ongoing natural pregnancy **56% (salpingectomy) vs 61% (salpingostomy)**, NS.
 - **DEMETER:** 2-year IUP **64% vs 70%**, NS.
- **Abnormal contralateral tube → salpingostomy** preferred if feasible.
- Irrigate and suction the pelvis to remove all trophoblast.

Persistent trophoblast

- Suspected with **stable or rising β -hCG** after surgery.
- No rupture → **single-dose MTX 50 mg/m²**. **Rupture/bleeding → second surgery.**

Medical versus surgical therapy

- **Multidose MTX vs laparoscopic salpingostomy** (RCT, n = 100): **no difference** in tubal preservation, primary success or fertility.
- **Single-dose MTX vs salpingostomy:** conflicting; long-term (8.6 y) success and spontaneous IUP rates similar.
- **Salpingectomy** has the highest resolution rate, but **future fertility and recurrence are similar** to MTX. Surgery, MTX and expectant care give similar subsequent IUP rates.
- **Equivalent outcomes** when: **stable, β -hCG <5000 mIU/mL, small mass, no cardiac activity**. MTX can still be offered to motivated women outside these limits who accept the risk of emergency surgery.

Expectant management

- For **select asymptomatic** women with a **very early tubal pregnancy and stable or falling β -hCG**, committed to follow-up and near emergency care. (Different from expectant care of a PUL.)
- Predictors of success: **low initial β -hCG**, a **significant fall over 48 h**, and an **inhomogeneous mass** (not a tubal ring or embryo).

Initial β -hCG	Spontaneous resolution
<175 mIU/mL	88–96%
<1000 mIU/mL	71–92%

- Later tubal patency and IUP rates are comparable to surgery.

Pearl: The three management options for tubal ectopics give similar future fertility. The choice rests on stability, β -hCG level, mass size, cardiac activity and whether the patient can be followed reliably.

5. Interstitial Pregnancy

Diagnosis

- Implants in the **tubal segment within the muscular uterine wall**. ("Cornual" pregnancy strictly means one in the **rudimentary horn** of a müllerian anomaly.)
- Specific risk factor: **prior ipsilateral salpingectomy**.
- **Ruptures later: after 8–16 weeks** of amenorrhea (myometrium allows more distention).
- Close to the uterine and ovarian arteries → **severe hemorrhage; mortality up to 2.5%**.
- **Sonographic criteria:**
 - **Empty uterine cavity**
 - Sac **separate from the endometrium** and **>1 cm from the most lateral edge** of the cavity
 - **Thin myometrial mantle <5 mm** around the sac
 - **Interstitial line sign:** an echogenic line from the sac to the endometrial cavity (**highly sensitive and specific**)
- Unclear cases: **3-D sonography, MRI or laparoscopy** (bulge **lateral to the round ligament** with normal distal tube and ovary).

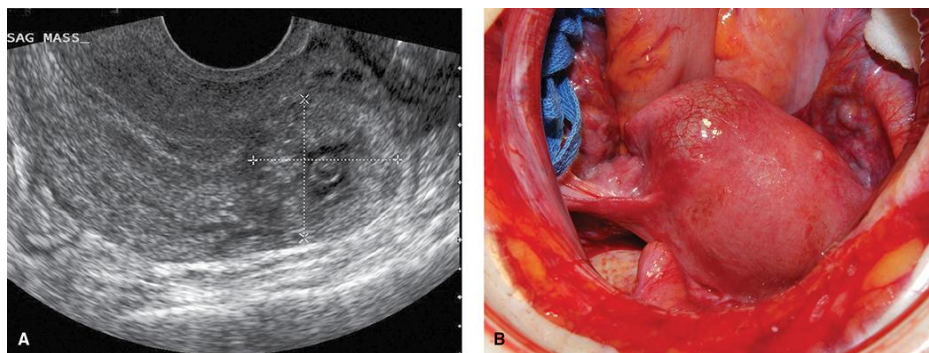


Figure 12-6. Interstitial pregnancy: (A) empty cavity with a mass cephalad and lateral to the fundus; (B) at laparotomy, the bulge lies lateral to the round ligament and medial to the isthmic tube.

Management

- **Surgical** (laparotomy or laparoscopy), with **intramyometrial vasopressin** to ↓ blood loss:
 - **Cornual resection:** wedge excision of the sac, surrounding myometrium and ipsilateral tube, **en bloc**
 - **Cornuostomy:** incise the myometrium and extract the pregnancy
 - Both need **layered myometrial closure**; follow **β-hCG** for remnant trophoblast.
- **Medical** (early diagnosis): no consensus. Systemic MTX **50 mg/m²** → **94%** success; multidose; **direct intrasac MTX**; **chemoembolization** (uterine artery MTX + UAE) + systemic MTX.
- Future rupture risk undefined → **elective cesarean at ≥37⁰/₇ weeks** (as after at-risk myomectomy).
- **Angular pregnancy** = eccentric implantation **within the cavity** near a cornu: **80%** reached viability without abnormal placentation or rupture → **managed as a normal pregnancy**.

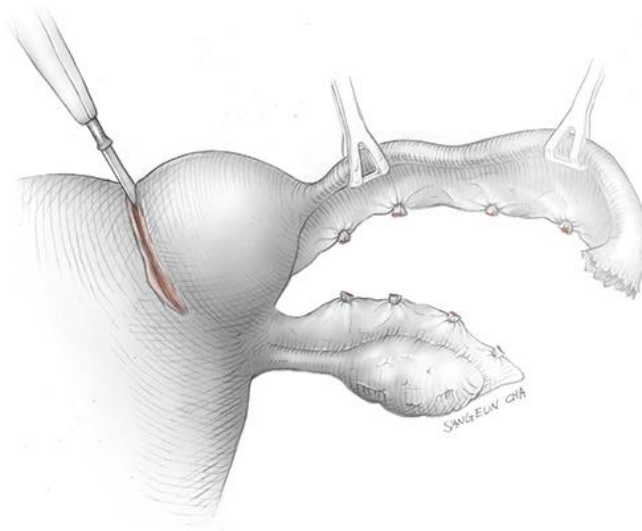


Figure 12-7. Cornual resection: wedge excision (angled inward) of the pregnancy, surrounding myometrium and ipsilateral tube en bloc, then layered closure.

6. Cesarean Scar Pregnancy

Diagnosis

- Implantation **within the myometrium of a prior hysterotomy scar**. Incidence **~1 in 2000** pregnancies, rising with cesarean rates.
- Usually present early with **pain and bleeding; up to 40% asymptomatic** (found on routine scan).
- **TVS criteria (Fig 12-8):**
 1. **Empty uterine cavity and empty endocervical canal**
 2. Placenta/sac **embedded in the hysterotomy scar niche**
 3. **Thin myometrial mantle** between sac and **bladder**
 4. **Prominent vascular pattern** at the scar
- **Endogenic** CSP (on the scar, grows toward the cavity) → variable outcomes. **Exogenic** (deep in the niche, grows toward the bladder) → **all needed hysterectomy with PAS** at delivery in one study.
- **Versus low IUP:** sac centre **below the midpoint** of the uterine length (cervix to fundus) → CSP.
- **Versus an aborting sac:** CSP shows **intense peritrophoblastic flow** on color Doppler and a **negative sliding sign** (does not move with probe pressure). An expelling sac is **avascular** and **slides**.
- MRI for inconclusive cases.

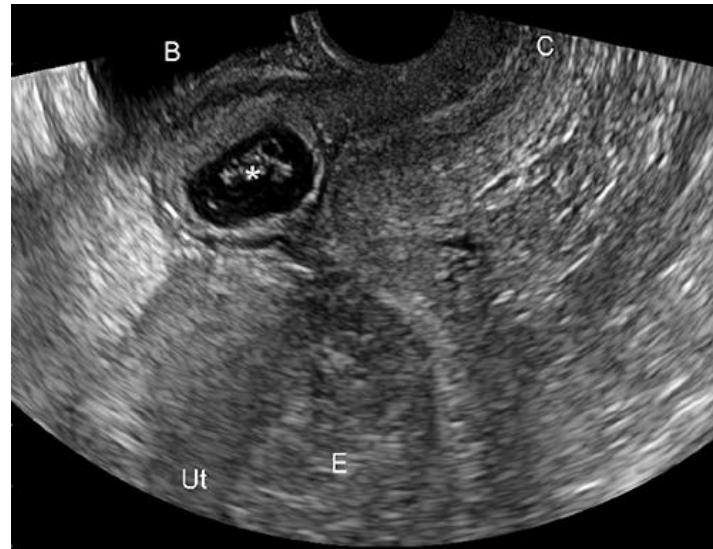


Figure 12-8. Cesarean scar pregnancy: empty cavity (E) and canal, sac (*) in the anterior scar niche, thin myometrium between the sac and bladder (B).

Management

- CSP may be a **precursor of placenta accreta spectrum**; many reach viability, with PAS complications. Rupture is less common than with typical ectopics.
- **Most successful operations:**
 1. **Laparoscopic isthmic resection**
 2. **Transvaginal isthmic resection** via anterior colpotomy
 3. **UAE then D & C** ± hysteroscopy
 4. **Hysteroscopic resection**
- SMFM: **ultrasound-guided vacuum aspiration** is suitable; **sharp curettage is not**. Hysterectomy if required or no further fertility wanted.
- **Medical:** slower and more variable. **Local intrasac MTX ~60%**; **local + systemic ~80%**. **SMFM recommends against systemic MTX alone.**
 - Local MTX **1 mg/kg or 50 mg**; in advanced gestations, first **KCl** into the sac (e.g. **1 mL of 2 mEq/mL**).
 - Bleeding → **Foley balloon** tamponade; **double-balloon catheter** + MTX has been used.
- After conservative treatment: good subsequent outcomes, but **PAS and recurrent CSP** (4 of 9 in one series) and **uterine AVMs** are risks.
- **Expectant management: SMFM recommends against it** (exception: early CSP with signs of failure).
 - Continuing CSPs: **uterine rupture/dehiscence 10%**; **hysterectomy 60% overall**; of 40 reaching the 3rd trimester, **17 had percreta** and 23 had hysterectomy.
 - Early CSPs **without cardiac activity: 70% uncomplicated miscarriage**, none needed hysterectomy.
 - If continued: **repeat cesarean at 34⁰/₇–35⁶/₇ weeks** after **betamethasone**.

7. Cervical Pregnancy

Diagnosis

- **Definition:** (1) **cervical glands opposite the placental attachment site** on histology, and (2) all or part of the placenta lies **below the entry of the uterine vessels** or below the anterior peritoneal reflection.
- Risk factors: **ART**, **prior uterine curettage**.
- **Painless vaginal bleeding in 90%** — may be severe.
- Exam: **distended, thin-walled cervix** with a partly open external os; a slightly enlarged fundus above it.

- **TVS findings (Fig 12-9):**
 1. **Hourglass uterus** with **ballooned cervix**
 2. Gestational tissue **at the level of the cervix**
 3. **No intrauterine gestational tissue**
 4. Part of the **endocervical canal between the gestation and the endometrial canal**
- Versus miscarriage in transit: **vascular** on color Doppler and a **negative sliding sign**. MRI/3-D TVS can help.



Figure 12-9. Cervical pregnancy: hourglass uterus with the gestational sac (arrow) low in a ballooned cervix and an empty uterine cavity above.

Management

- **MTX is first-line** in stable women (Parkland): single- or multidose systemic (Table 12-1), or **50 mg directly into the sac**, or chemoembolization.
- **<12 weeks: 91% resolution with uterine preservation.**
- **Higher MTX failure if:** GA >9 weeks, β -hCG >10,000 mIU/mL, CRL >10 mm, cardiac activity → add **feticidal KCl** into the fetus/sac.
- Sonographic resolution **lags far behind** β -hCG decline.
- **Suction evacuation** (reduce bleeding with preoperative **UAE**, **intracervical vasopressin**, a **cerclage at the internal os**, or **hemostatic sutures at 3 and 9 o'clock** to ligate the cervical branches of the uterine artery).
- **Hysterectomy** for uncontrolled bleeding (**ureters** close to the ballooned cervix → urinary injury risk).
- **Hemorrhage: 26F Foley with a 30-mL balloon** in the cervix for tamponade, left **24–48 h** and deflated gradually over a few days. UAE for bleeding or prevention.

8. Abdominal Pregnancy

Diagnosis

- Implantation in the **peritoneal cavity**, excluding tubal, ovarian and intraligamentous sites. Most follow **early tubal rupture or tubal abortion with reimplantation**.
- Symptoms absent or vague; **MSAFP may be raised**. Later: **abnormal fetal lie, displaced cervix**.
- **Sonographic clues:**
 - Fetus or placenta **eccentric in the pelvis or separate from the uterus**
 - **No myometrium** between the fetus and the maternal abdominal wall or bladder
 - **Bowel loops around the sac**
 - **Oligohydramnios** (common, nonspecific)

- **MRI** often needed (also maps the placenta).

Management

- Expectant care risks **sudden dangerous hemorrhage; fetal malformations/deformations in 20%** → **termination is generally indicated** once diagnosed, and rarely justified to wait **before 24 weeks** (some wait for viability with close surveillance).
- **Surgery:** deliver the fetus; assess the placental implantation **without provoking hemorrhage**; avoid unnecessary exploration.
- ⚠ **Placental removal can cause torrential bleeding** (no myometrial contraction to close vessels).
 - Remove it only if clearly safe or already bleeding; **ligate its feeding vessels first**; dilute vasopressin for early gestations.
- **Leaving the placenta in situ** (lesser evil): ↓ immediate hemorrhage but risks **abscess, adhesions, bowel/ureteral obstruction, wound dehiscence** (often requiring later removal).
 - Monitor involution with **β-hCG, color Doppler or MRI**; resorption can take **months to years**.
 - **Postoperative MTX** hastens involution but causes necrosis → **infection/abscess**. Placental embolization may help.

9. Ovarian Pregnancy

- **Spiegelberg criteria (1878):**
 1. **Ipsilateral tube intact** and separate from the ovary
 2. Ectopic pregnancy **occupies the ovary**
 3. Connected to the uterus by the **utero-ovarian ligament**
 4. **Ovarian tissue** shown histologically **amid placental tissue**
- Risk factors like tubal, but **ART and IUD failure** are prominent.
- Presentation mirrors tubal ectopic; **early rupture** is usual.
- **TVS:** anechoic centre within a **wide echogenic ring**, surrounded by **ovarian cortex**. Often not diagnosed until surgery; may be mistaken for a **hemorrhagic corpus luteum**.
- **Surgical management:** small → **ovarian wedge resection or cystectomy**; large → **oophorectomy**. Follow β-hCG after conservative surgery.

Memory hook (Spiegelberg): "Tube spared, ovary occupied, ligament linked, ovarian tissue on histology."

10. Heterotopic Pregnancy

- IUP + ectopic pregnancy together; most often **IUP + ampullary** tubal.
- **Natural incidence ~1 in 30,000**; with **ART ~9 in 10,000**.
- Symptoms come from the ectopic. **Rupture is more common** because the visible IUP falsely reassures and the ectopic is missed.
- **Hemorrhage → surgery** (resection or suction aspiration, depending on site). **UAE and vasopressin are less desirable** (↓ uterine flow to the IUP).
- Ovarian heterotopic with early **corpus luteum excision → progesterone** supplementation.
- **No significant bleeding:** inject the ectopic sac with **KCl or hyperosmolar glucose** ± later aspiration.
- ⚠ **MTX is avoided** (toxic to the IUP).
- Neonatal outcomes of the IUP are not appreciably worse; early growth scans are reasonable. **Delivery route** depends on myometrial integrity after treatment.

Quick-Recall: Ectopic Sites at a Glance

Site	Frequency / incidence	Key diagnostic feature	First-line management
Ampullary / tubal	95% of ectopics (ampulla 70%)	Adnexal mass separate from ovary; ring of fire	MTX if stable and suitable; else laparoscopic salpingectomy/salpingostomy
Interstitial	2% of tubal	Sac >1 cm from cavity edge, mantle <5 mm, interstitial line sign	Cornual resection / cornuostomy; MTX if early; cesarean ≥37 wk later
Cesarean scar	1 in 2000	Sac in scar niche, thin mantle to bladder, negative sliding sign	Resection (laparoscopic, vaginal, hysteroscopic), UAE + D & C, or US-guided aspiration; local ± systemic MTX (not systemic alone)
Cervical	Rare	Hourglass uterus, ballooned cervix, painless bleeding in 90%	MTX (systemic or intrasac) ± KCl; suction ± UAE/cerclage/3 and 9 o'clock sutures; Foley tamponade
Abdominal	Rare	Fetus outside uterus, no myometrium, bowel around sac, MRI	Laparotomy; deliver fetus; leave placenta if removal unsafe
Ovarian	Rare	Spiegelberg criteria; ring within ovarian cortex	Wedge resection/cystectomy or oophorectomy
Heterotopic	1/30,000 (9/10,000 with ART)	IUP plus adnexal mass	Surgery if bleeding; else KCl or hyperosmolar glucose into sac; no MTX

Key Numbers to Memorize

- Ectopic rate **1.4–1.5%**; **3%** of pregnancy-related deaths; tubal **95%**; ampulla **70%**, isthmus **12%**, fimbria **11%**, interstitial **2%**
- Recurrence after one ectopic **~10%**; after salpingitis **up to 9%**
- Anti-D **50–120 µg** (1st trimester), **300 µg** later; urine hCG detection **20–25**, serum **≤5 mIU/mL**; discriminatory zone **1500–2000** (sac in 99% >**3510**)
- Normal IUP 48-h rise **≥53%** (≥35% conservative); **1/3** of ectopics also rise **≥53%**
- Trilaminar endometrium: specificity **94%**, sensitivity **38%**; stripe **<8 mm** → no normal IUP
- Adnexal: complex mass **60%**, ring **20%**, yolk sac/embryo **13%**
- Morison pouch fluid at **400–600 mL**; leukocytosis up to **30,000/µL**
- Progesterone **<6 ng/mL** nonviable; **>25 ng/mL** live IUP
- Presumed ectopic wrong in **27%** without curettage; frozen section **95%** accurate
- MTX single dose **50 mg/m²**; check hCG day **4 and 7**, need **≥15%** fall; second dose in **20%**; multidose MTX **1 mg/kg** + leucovorin **0.1 mg/kg**, max **4 doses**
- MTX success **~90%**; failure **1.5%** (<1000) → **14.3%** (5000–10,000); mass **≤3.5 cm** → **93%**; cardiac activity → **87%**
- Separation pain **65–75%**; resolution mean **34 days** (max 109); wait **3 months** to conceive
- Salpingostomy incision **10–15 mm**; persistent trophoblast **5–15%**; ESEP **56 vs 61%**; DEMETER **64 vs 70%**
- Equivalent medical vs surgical: β-hCG **<5000**, small, no cardiac activity
- Expectant: hCG **<175** → **88–96%**; **<1000** → **71–92%**
- Interstitial: rupture at **8–16 wk**; mortality **2.5%**; mantle **<5 mm**, **>1 cm** from cavity; MTX **94%**; cesarean **≥37 wk**
- CSP **1/2000**; asymptomatic **40%**; expectant → hysterectomy **60%**, rupture **10%**; deliver **34–35th wk**
- Cervical: painless bleeding **90%**; MTX **91%** (<12 wk); failure if **>9 wk**, hCG **>10,000**, CRL **>10 mm**, cardiac activity; Foley **26F / 30 mL** for **24–48 h**
- Abdominal: malformation/deformation **20%**, don't wait before **24 wk**; heterotopic **1/30,000** (ART **9/10,000**)